

Education

University of Denver	Expected 2027
M.S. in Applied Mathematics (Applied Statistics Focus)	GPA: 4.0
Metropolitan State University of Denver	
B.S. in Computer Science	
B.S. in Data Science and Machine Learning	

Core Competencies

Languages & Frameworks: Python (NumPy, pandas, Seaborn), R (tidyverse, GGplot), SQL
ML & AI: scikit-learn, PyTorch, TensorFlow, Keras, Hugging Face, CNNs (image/audio), LLM integration (Copilot, Claude), prompt engineering, agentic system design.
Data & MLOps: ETL pipelines, REST APIs, Docker, AWS, Terraform, workflow automation, reproducible pipeline development, technical documentation, data visualization.
Statistics: regression, classification, clustering, ANOVA, correlation, causality, probability theory, statistical inference

Professional Experience

University of Colorado Anschutz Medical Campus	Aurora, CO
Machine Learning Researcher (ML / Bioinformatics)	Dec 2025 – Present
- Engineered a dynamic data-fetching and preprocessing pipeline for tuberculosis drug repurposing, improving scalability across diverse genomic datasets. - Led comparative transcriptomic analyses (DESeq2, clusterProfiler) to identify disease-associated pathways and networks; preparing second-author findings for publication. Preprint: bioRxiv 2025.06.02.657296v2 - Built and evaluated ML models for disease severity and phenotype prediction, including tree-based methods and SVMs on 100,000+ sample-scale datasets. - Translated complex biological questions into computational workflows, ensuring rigorous data cleaning, visualization, and reproducible analysis.	
DeNOVO Solutions (Capstone)	Aurora, CO
Full-Stack Engineer – MLOps Drift Detection	Aug 2025 – Dec 2025
- Built a full-stack drift detection application using React and Django, designed for integration into production ML services. - Implemented REST APIs and an ETL pipeline with automated notifications to flag data integrity issues and model drift in real time. - Designed dashboards to visualize model performance metrics and feature distributions for ML stakeholders.	
Metropolitan State University of Denver	Denver, CO
Teaching Assistant / CS Tutor (ML, CS I & II, Web App Dev)	2024 – 2025
- Mentored students in Machine Learning and Web Application Development, emphasizing deployment-oriented thinking and practical model evaluation. - Guided 150+ students in core CS topics (OOP, data structures) and DS/ML workflows using Python, scikit-learn, and modern tooling. - Served as a CS Department Ambassador at the Denver AI Summit (DenAI), facilitating discussions on democratizing AI and human-centered applications.	

Selected Projects & Research

Human-Centered Pedagogy Agents	Hugging Face, Ollama
Designed LLM-based agent prototypes to provide equitable, context-aware support for educators, focusing on workflow automation and explainable recommendations.	
Deep Learning Classification (Image & Audio)	Pytorch, numpy, pandas
Built CNN-based classifiers for handwritten digits and musical instruments, including custom ETL pipelines (e.g., YouTube audio segment extraction) and GPU-accelerated training.	
BrainBattles – Multiplayer Trivia Platform	Django, React, PostgreSQL
- Implemented core backend logic and REST APIs in Django, including game session management, scoring, and question delivery. - Contributed to front-end components in React and helped deploy the application, maintaining high test coverage and responsive performance.	